

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET G — FINAL MASTERY & GRAND CHALLENGE TEST (60 MINS)

CHAPTER 04: DESCRIBING MOTION & KINEMATICS

TIME ALLOWED: 60 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-G-04

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET G - FINAL MASTERY & GRAND CHALLENGE TEST):

1. The question paper comprises **25 High-Tier Mastery & Olympiad-Standard Multiple Choice Questions** of **1 mark each**.
2. Tests **In-Depth Conceptual Mastery, Multi-Step Numericals & Rigorous Scientific Deductions**: Recommended target time is **60 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.



GRAND CHALLENGE TIER: This paper features comprehensive multi-step numericals, advanced Olympiad/NTSE-aligned scenarios, and rigorous experimental synthesis designed to achieve absolute 100/100 mastery.

Q.1 A particle moves along a circular path of radius R . What is the ratio of distance to displacement magnitude after completing a semi-circle (180°)?

- (A) $\pi : 2$ (B) $2 : \pi$
(C) $\pi : 1$ (D) $1 : 1$

Q.2 A stone dropped from the top of a tower of height H reaches the ground in time t . Where is the stone at time $t/2$?

- (A) At height $H/2$ (B) At height $3H/4$ above ground
(C) At height $H/4$ above ground (D) At ground level

Q.3 A car accelerates uniformly from rest at 2 m/s^2 for 10 s , then coasts at constant speed for 20 s , and finally brakes to rest in 5 s . Total distance covered is:

- (A) 100 m (B) 550 m
(C) 600 m (D) 450 m

Q.4 A body starts from rest and moves with uniform acceleration. The ratio of distances covered in the 1st, 2nd, and 3rd seconds of motion is:

- (A) $1 : 2 : 3$ (B) $1 : 4 : 9$
(C) $1 : 3 : 5$ (Galileo's odd numbers law) (D) $1 : 1 : 1$

Q.5 If the displacement of a body is proportional to the square of time ($s \propto t^2$), the body is moving with:

- (A) Uniform velocity (B) Uniform acceleration
(C) Increasing acceleration (D) Zero velocity

Q.6 A bullet moving at 200 m/s penetrates a wooden block to a depth of 10 cm before stopping. What is its average deceleration inside the block?

- (A) $2 \times 10^5 \text{ m/s}^2$ (B) $4 \times 10^5 \text{ m/s}^2$
(C) $2 \times 10^4 \text{ m/s}^2$ (D) 10^5 m/s^2

Q.7 A wheel of radius 0.5 m rotates at 120 revolutions per minute (rpm). What is the linear speed of a point on the rim?

- (A) $\pi\text{ m/s} \approx 3.14\text{ m/s}$ (B) $2\pi\text{ m/s} \approx 6.28\text{ m/s}$
 (C) 60 m/s (D) 120 m/s

Q.8 Two trains of length 150 m each travel on parallel tracks in opposite directions with speeds 36 km/h and 54 km/h . Time taken to completely cross each other is:

- (A) 12 s (B) 15 s
 (C) 20 s (D) 30 s

Q.9 An object thrown vertically upward reaches a peak height H . What is its speed at half the peak height ($H/2$)?

- (A) $u/2$ (B) $u/\sqrt{2}$
 (C) $\sqrt{2} u$ (D) $u/4$

Q.10 Can a body have a constant speed and a changing velocity simultaneously?

- (A) Yes, in uniform circular motion (B) No, if speed is constant, velocity must be constant
 (C) Only in deep space (D) Only when acceleration is zero

Q.11 What does a negative slope on a velocity-time graph indicate?

- (A) Increasing acceleration (B) Retardation (deceleration)
 (C) Zero velocity (D) Infinite speed

Q.12 The displacement s of a particle varies with time as $s = 3t + 2t^2$. Its initial velocity ($t=0$) and acceleration are:

- (A) $u = 3\text{ m/s}$, $a = 4\text{ m/s}^2$ (B) $u = 0$, $a = 2\text{ m/s}^2$
 (C) $u = 2\text{ m/s}$, $a = 3\text{ m/s}^2$ (D) $u = 3\text{ m/s}$, $a = 2\text{ m/s}^2$

Q.13 If a body covers distance x in time t such that $x = at^3$, its acceleration is:

- (A) Constant (B) Decreasing
 (C) Proportional to time ($a \propto t$) (D) Zero

Q.14 A ball dropped from height h bounces back to height $h/4$. What is the coefficient of restitution e ?

- (A) 0.25 (B) 0.50
 (C) 0.75 (D) 0.10

Q.15 A particle moves with constant acceleration $a = 2\text{ m/s}^2$. If its velocity after 5 seconds is 15 m/s, what was its initial velocity?

- (A) 5 m/s (B) 10 m/s
 (C) 25 m/s (D) 0 m/s

Q.16 Under what condition is the magnitude of average velocity equal to average speed?

- (A) When the body moves along a straight line in a fixed single direction without reversing (B) When the body moves in a closed circle
 (C) When acceleration is zero only (D) Never

Q.17 What is the centripetal acceleration of an Earth satellite orbiting at speed $v = 8000\text{ m/s}$ at orbital radius $r = 6.4 \times 10^6\text{ m}$?

- (A) 10.0 m/s^2 (B) 9.8 m/s^2
 (C) 1.0 m/s^2 (D) 0 m/s^2

Q.18 A car travelling at speed v stops in distance d when braking force F is applied. If the car's initial speed is $3v$, what is the new stopping distance under the same braking force?

- (A) $3d$ (B) $6d$
(C) $9d$ (D) $d/3$

Q.19 A body is projected vertically upwards with velocity u . What is its velocity when it returns to the point of projection (neglecting air resistance)?

- (A) $+u$ (B) $-u$ (same magnitude, opposite direction)
(C) Zero (D) $u/2$

Q.20 What is the ratio of average speed to average velocity for a body that travels along the circumference of a circle and returns to the start?

- (A) 1 (B) Zero
(C) Undefined (division by zero) (D) π

Q.21 A particle travelling with uniform acceleration covers 20 m in the 2nd second and 30 m in the 3rd second. What is its acceleration?

- (A) 5 m/s^2 (B) 10 m/s^2
(C) 2.5 m/s^2 (D) 15 m/s^2

Q.22 What is the instantaneous velocity of a body thrown vertically upwards when it reaches the peak of its trajectory?

- (A) 9.8 m/s (B) Zero
(C) Infinitely large (D) $u/2$

Q.23 What is the acceleration of the same body at the peak of its trajectory?

- (A) Zero (B) 9.8 m/s^2 downward
(C) 9.8 m/s^2 upward (D) Undefined

Q.24 A graph of v^2 versus s for a particle starting from rest and moving with uniform acceleration is a:

- (A) Straight line passing through the origin with slope $2a$ (B) Parabola curving upwards
(C) Hyperbola (D) Horizontal line

Q.25 Which property of motion is strictly relative and depends on the observer's frame of reference?

- (A) Rest and motion (B) Velocity
(C) Displacement (D) All of the above

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET G)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & STEP-BY-STEP EXPLANATIONS (SET G)

Q. No.	Correct	Rigorous Scientific Solution & Formula Derivation
Q.1	(A)	Distance along semicircle $s = \pi R$. Displacement across diameter $s = 2R$. Ratio $s = \frac{\pi R}{2R} = \frac{\pi}{2}$.
Q.2	(B)	Distance fallen in $t/2$: $s = \frac{1}{2}gt/2^2 = \frac{1}{4}(\frac{1}{2}gt^2) = \frac{1}{4}H$. Height remaining above ground $s = H - \frac{1}{4}H = \frac{3}{4}H$.
Q.3	(B)	$v_{\max} = 2 \times 10 = 20 \text{ m/s}$. Phase 1: $s_1 = \frac{1}{2}(2)(10)^2 = 100 \text{ m}$. Phase 2: $s_2 = 20 \times 20 = 400 \text{ m}$. Phase 3: $s_3 = \frac{1}{2}(20)(5) = 50 \text{ m}$. Total $s = 100 + 400 + 50 = 550 \text{ m}$.
Q.4	(C)	Distance in n -th second $s_n = u + \frac{1}{2}a(2n - 1)$. For $u = 0$, $s_n \propto (2n - 1)$, giving ratios $1 : 3 : 5 : 7 \dots$.
Q.5	(B)	Since $s = \frac{1}{2}at^2$, $s \propto t^2$ implies that acceleration a is strictly constant.
Q.6	(A)	$v^2 = u^2 + 2as \rightarrow 0 = (200)^2 + 2a(0.10) \rightarrow 0.20a = -40000 \rightarrow a = -200,000 \text{ m/s}^2 = -2 \times 10^5 \text{ m/s}^2$.
Q.7	(B)	Frequency $f = 120/60 = 2 \text{ rev/s}$. Speed $v = 2\pi r f = 2\pi(0.5)(2) = 2\pi \approx 6.28 \text{ m/s}$.
Q.8	(A)	Relative speed $s = 36 + 54 = 90 \text{ km/h} = 90 \times (5/18) = 25 \text{ m/s}$. Total distance to cross $s = 150 + 150 = 300 \text{ m}$. Time $s = 300/25 = 12 \text{ s}$.
Q.9	(B)	$v^2 = u^2 - 2g(H/2) = u^2 - gH$. Since $H = \frac{u^2}{2g}$, $gH = \frac{u^2}{2}$. Thus $v^2 = u^2 - \frac{u^2}{2} = \frac{u^2}{2} \rightarrow v = \frac{u}{\sqrt{2}}$.
Q.10	(A)	In uniform circular motion, speed is constant while the direction of motion continuously turns, continuously altering velocity.
Q.11	(B)	A negative slope $\Delta v / \Delta t < 0$ means velocity is decreasing with time, representing retardation.
Q.12	(A)	Comparing with $s = ut + \frac{1}{2}at^2$: $u = 3 \text{ m/s}$, and $\frac{1}{2}a = 2 \rightarrow a = 4 \text{ m/s}^2$.
Q.13	(C)	$v = \frac{dx}{dt} = 3at^2$, and $a_{\text{acc}} = \frac{dv}{dt} = 6at \propto t$. Acceleration increases linearly with time.
Q.14	(B)	Rebound height $h' = e^2 h$. Here $h/4 = e^2 h \rightarrow e^2 = 1/4 \rightarrow e = 0.50$.
Q.15	(A)	$v = u + at \rightarrow 15 = u + (2 \times 5) \rightarrow u = 15 - 10 = 5 \text{ m/s}$.
Q.16	(A)	Average velocity equals average speed only when total distance equals displacement magnitude (unidirectional straight-line path).
Q.17	(A)	$a_c = \frac{v^2}{r} = \frac{(8000)^2}{6.4 \times 10^6} = \frac{64 \times 10^6}{6.4 \times 10^6} = 10.0 \text{ m/s}^2$.
Q.18	(C)	Stopping distance $s \propto v^2$. If speed triples ($3v$), braking distance increases by $3^2 = 9$ times ($9s$).
Q.19	(B)	By conservation of mechanical energy, impact speed equals projection speed, directed downward ($-u$).
Q.20	(C)	Net displacement is zero; average velocity is 0. Dividing non-zero average speed by 0 is mathematically undefined.
Q.21	(B)	$s_n = u + a(n - 0.5)$. Difference $s_3 - s_2 = a \rightarrow 30 - 20 = 10 \text{ m/s}^2$.
Q.22	(B)	At the peak apex, the object momentarily halts ($v=0$) before reversing its vertical trajectory.
Q.23	(B)	Even though instantaneous velocity is zero, gravitational acceleration $g = 9.8 \text{ m/s}^2$ downward acts continuously.

Q. No.	Correct	Rigorous Scientific Solution & Formula Derivation
Q.24	(A)	From $v^2 = 2as$, plotting v^2 on y-axis against s on x-axis produces a straight line $y = mx$ with slope $m = 2a$.
Q.25	(D)	Rest, motion, displacement, and velocity are all defined relative to a specified frame of reference.